

Discussion Assignment 6

Cruise Control Controls

MATH 2210, Spring 2018

Background

Pontryagin's maximum principle is used in a field of mathematics and engineering called *optimal control theory* to find the best possible control for taking a dynamical system from one state to another, especially in the presence of constraints for the state or input controls. In plain English, Pontryagin's maximum principle tells you the best way to 'steer' something governed by predetermined dynamics (usually just the laws of physics, in some way) from one 'trajectory' (or path) to another.

For example, say you have a self-driving car transporting a passenger to work when it encounters a fender-bender that has caused a minor traffic jam. While there might be many other routes the self-driving car can take to avoid the accident, its programmer wants to make sure that the new trajectory the computer 'steers' the car to is optimal in some sense – probably something along the lines of "the car must obey traffic laws when changing course" and "changing to the new course shouldn't take longer than waiting for traffic to clear up."

In symbols, we often represent control systems by an equation like

$$\frac{d}{dt}\phi(t) = f(\phi(t), u(t))$$

where $\phi(t)$ is a function of time that outputs the state of the system (usually a vector in many dimensions) at a given time, $u(t)$ is the control (i.e. "press the pedal and turn the wheel this much at time t "), and f is a function which describes the dynamics of the system (i.e. f contains all the 'rules' the system must follow, like obeying gravity or something similar). In general, we consider the trajectory $\phi(t)$ to be the 'solution' to the system given above – that is, if we decide on a control and use it, the laws of physics do what they do and we end up with a trajectory satisfying the dynamics of the system.

In this context, the problem Pontryagin's maximum principle addresses would be something like "find a control $u(t)$ such that the resulting trajectory $\phi(t)$ minimizes the cost function M ." In our self-driving car example, M would be a function which measures how long the trip took and how many traffic laws the car broke. We would then say that our control was 'optimal' if the resultant trajectory broke as few laws as possible and took the least time (i.e. minimized M). Calculating optimal controls is usually exceedingly difficult, but is still often possible – you can find examples in things as simple as your thermostat, or in things as complicated as maneuvering a high-precision missile launched from a predator drone traveling several thousand miles an hour.

Pontryagin's maximum principle is more complicated than what we're ready to handle in this class, but it does heavily rely on the concept of Lagrange multipliers. In this problem, suppose we're designing a self-driving car and want it to be able to handle the situation given in the example above. Doing away with the control system, suppose we already know our travel time is given by the function

$$f(x, y) = x^2 y$$

where x measures the time spent in turns (in seconds) and y is the average acceleration of the vehicle (in m/s^2). Simplifying things further, suppose our cost function just describes constraints on the distance our gas tank will allow us to travel and is given by the function.

$$c(x, y) = x^2 + y^2 - 3.$$

Suppose we want to see how fast the car can get to its destination using all of the gas in the tank (so that we don't end up re-route the car when we can't transport our passenger to their destination in the first place). To do this, you need to solve the following problem:

Maximize $f(x, y) = x^2 y$ subject to the constraint $c(x, y) = x^2 + y^2 - 3 = 0$.

Using Lagrange multipliers, what values of x and y yield the global maximum of f subject to the constraint $c(x, y) = 0$? Assume that our time spent in turns and our average acceleration can't be negative for simplicity (that way, while your calculations may produce multiple answers, you should only end up with one after eliminating options with negative values of x and y). Choose the best answer from the options below.

- (a) $x = \sqrt{2}, y = 1$
- (b) $x = 0, y = \sqrt{3}$
- (c) $x = 2, y = 1$
- (d) $x = \sqrt{3}, y = \sqrt{5}$

Group Work

The group work portion of your grade is worth 3 points. As a group you are allowed to submit three answers. However, incorrect answers will be counted as wrong. For example, if the correct answer is (d) and your group submits $(a), (d), (d)$ as their answers the score will be $2/3$ on this portion of the assignment.

Individual Work

Now you are to write a report, one page maximum, that convinces an audience of your peers you have correctly determined the solution to the problem above. This report will need to be saved as a pdf and submitted to your discussion sections WyoCourses page. A scoring rubric and guideline on writing mathematics properly is posted on the WyoCourses page.